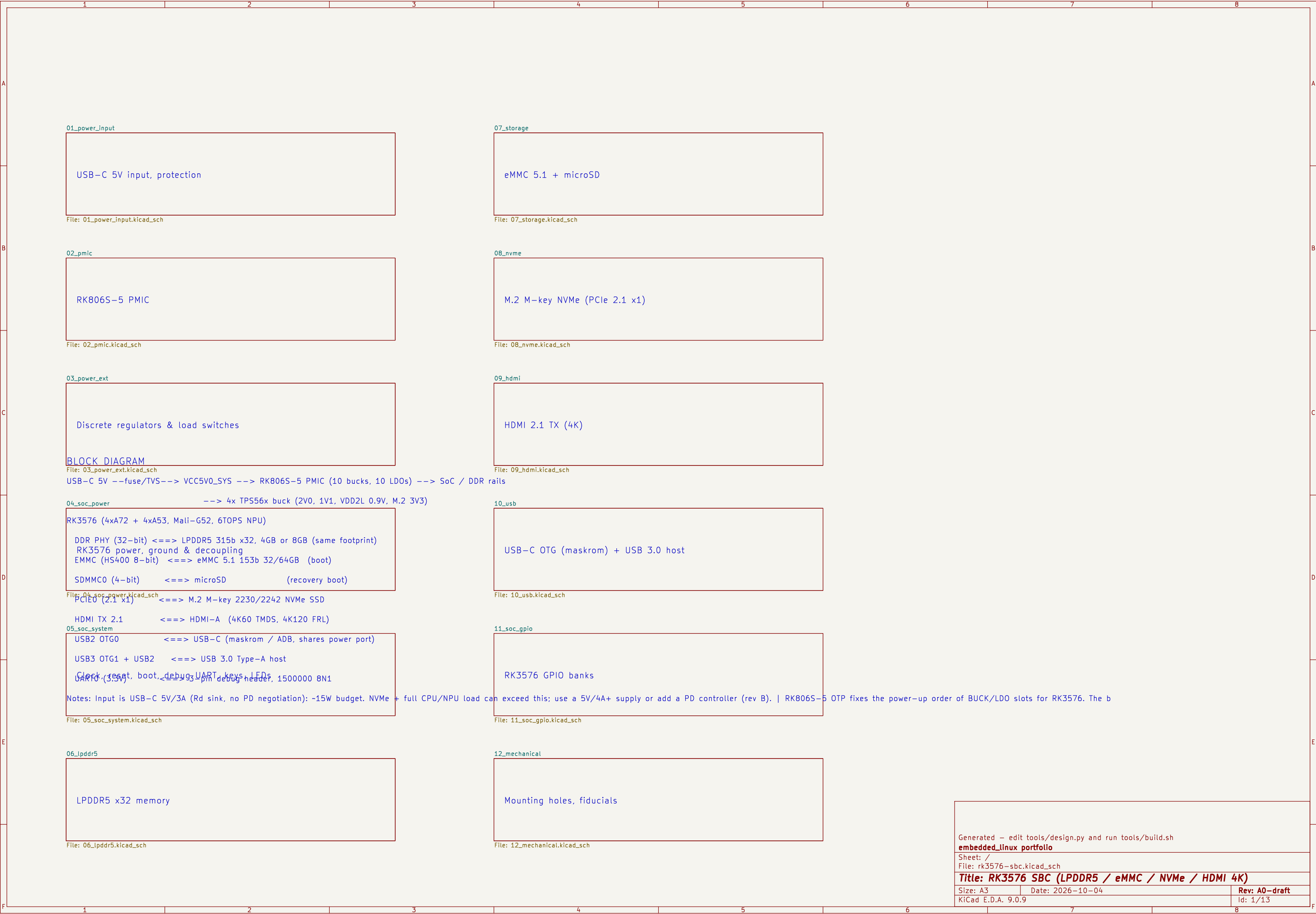
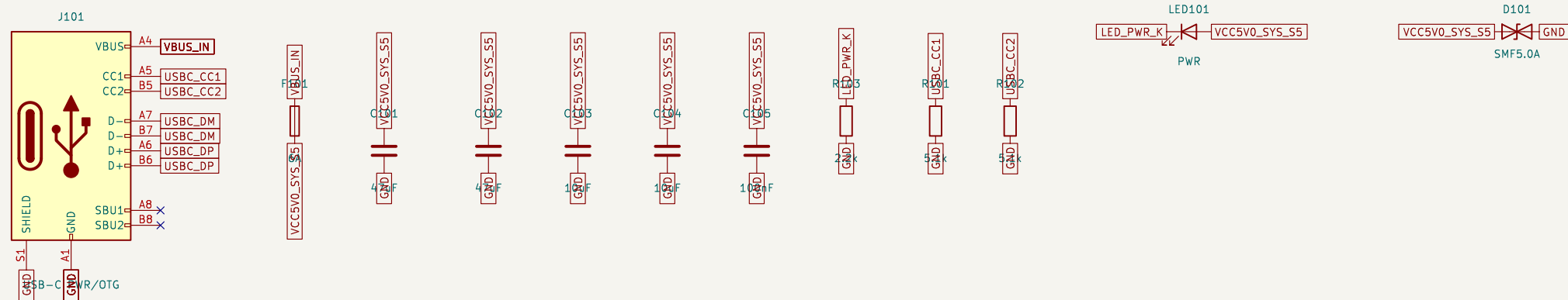
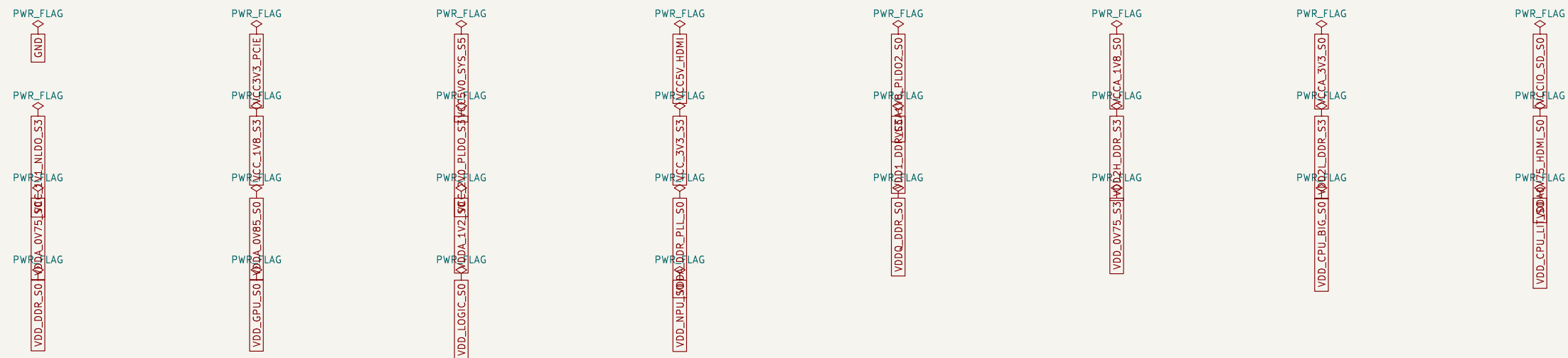




01_power_input: USB-C 5V input, protection

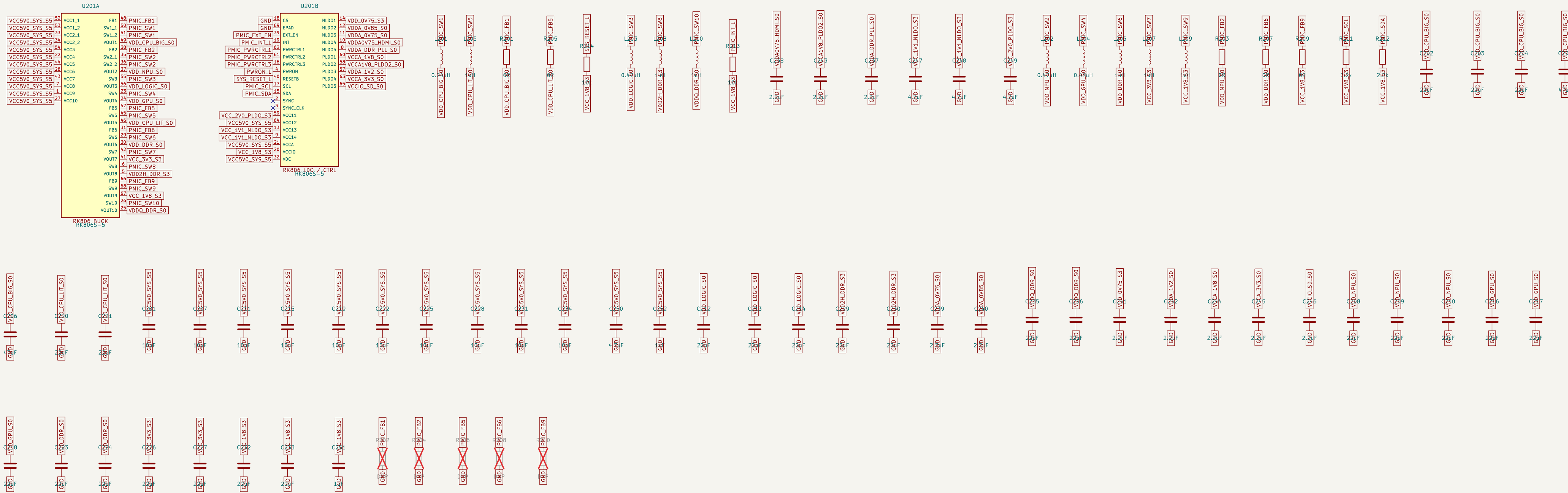


ERC power flags: rails generated by PMIC/regulator pins typed passive

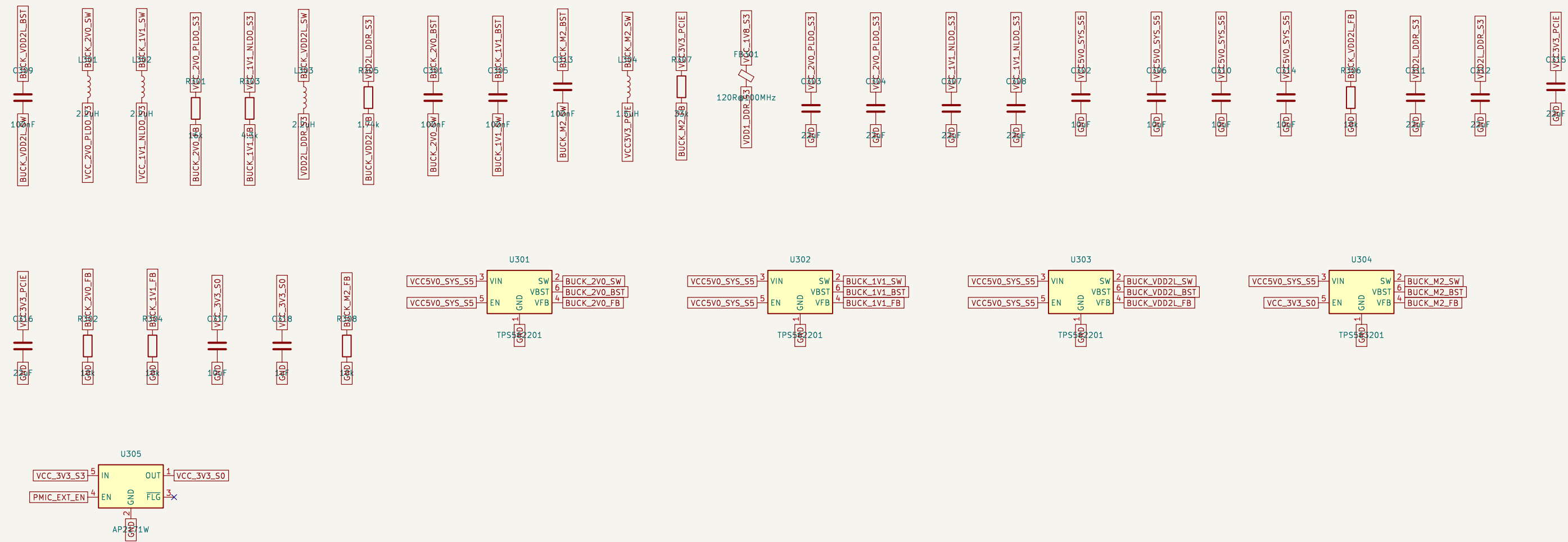


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RK3576 SBC (LPDDR5 / eMMC / NVMe / HDMI 4K)		
Sheet: /01_power_input/		
File: 01_power_input.kicad_sch		
Title: USB–C 5V input, protection		
Size: A3	Date: 2026–10–04	Rev: A0–draft
KiCad E.D.A. 9.0.9		Id: 2/13

02_pmic: RK806S–5 PMIC

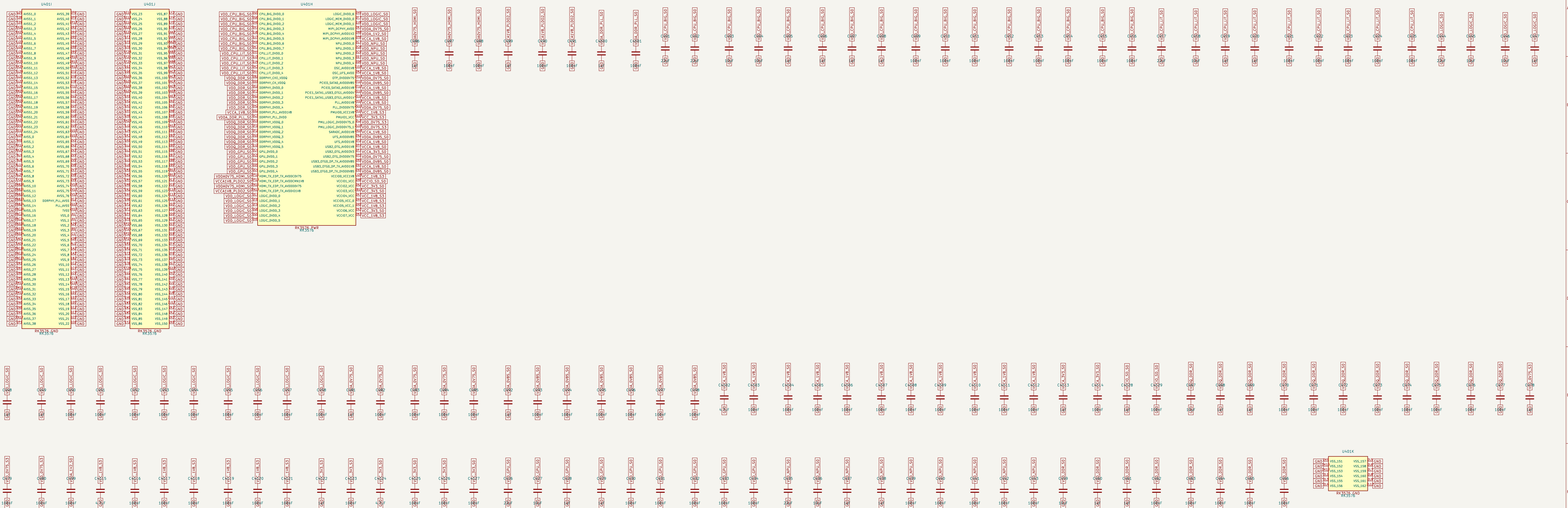


03_power_ext: Discrete regulators & load switches

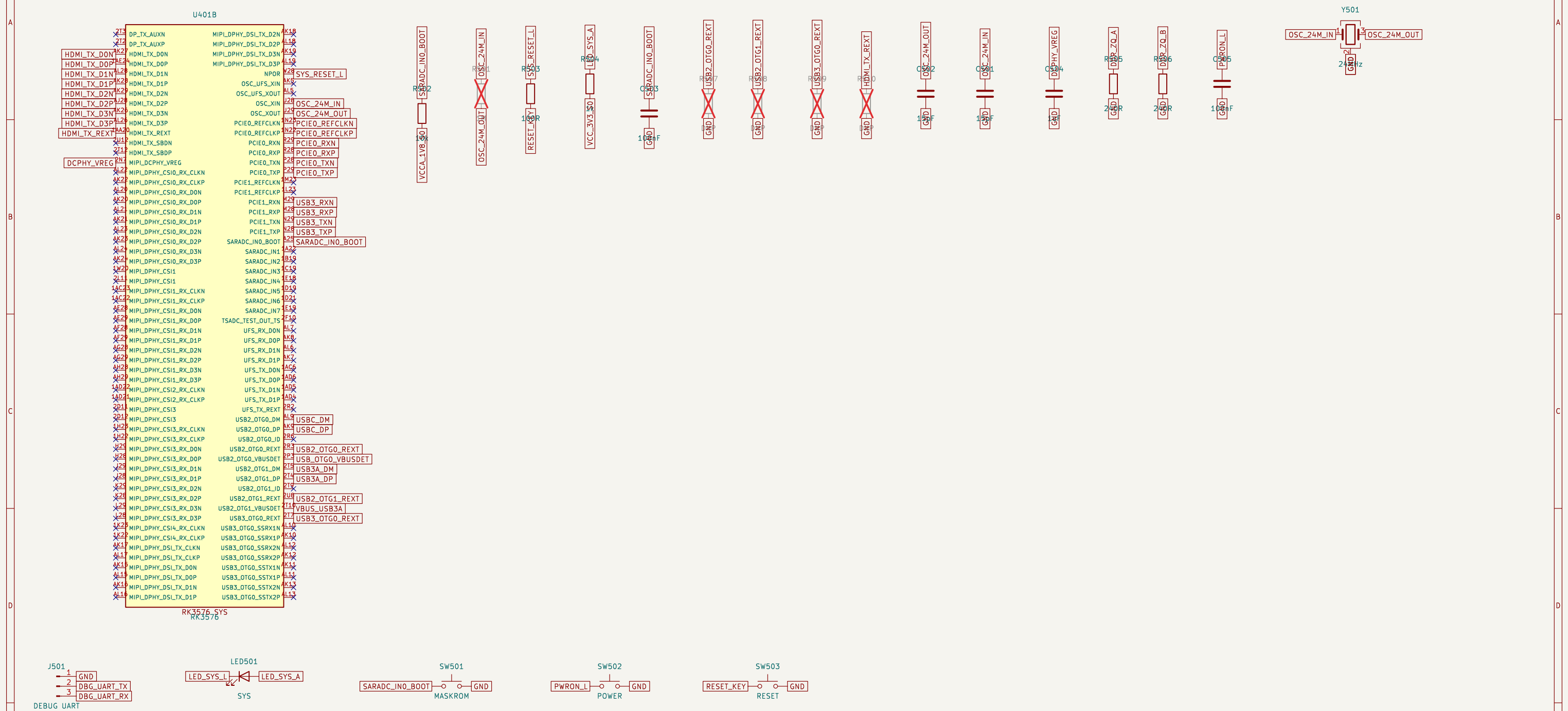


Generated by tools/gen_schematic.py from tools/design.py – edit the source, not this file		
RK3576 SBC (LPDDR5 / eMMC / NVMe / HDMI 4K)		
Sheet: /03_power_ext/		
File: 03_power_ext.kicad_sch		
Title: Discrete regulators & load switches		
Size: A3	Date: 2026-10-04	Rev: A0-draft
KiCad E.D.A. 9.0.9		Id: 4/13

04_soc_power: RK3576 power, ground & decoupling



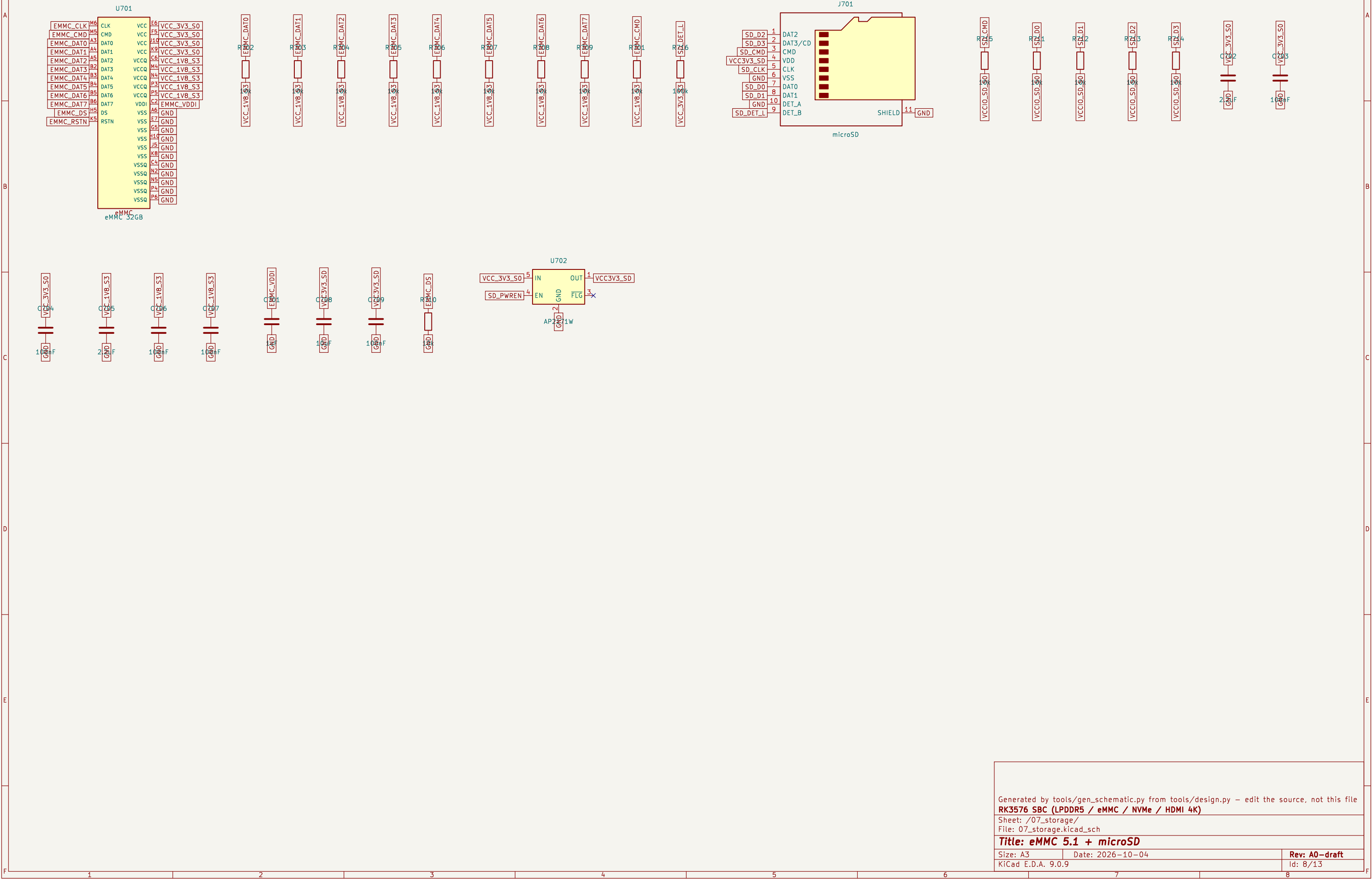
05_soc_system: Clock, reset, boot, debug UART, keys, LEDs



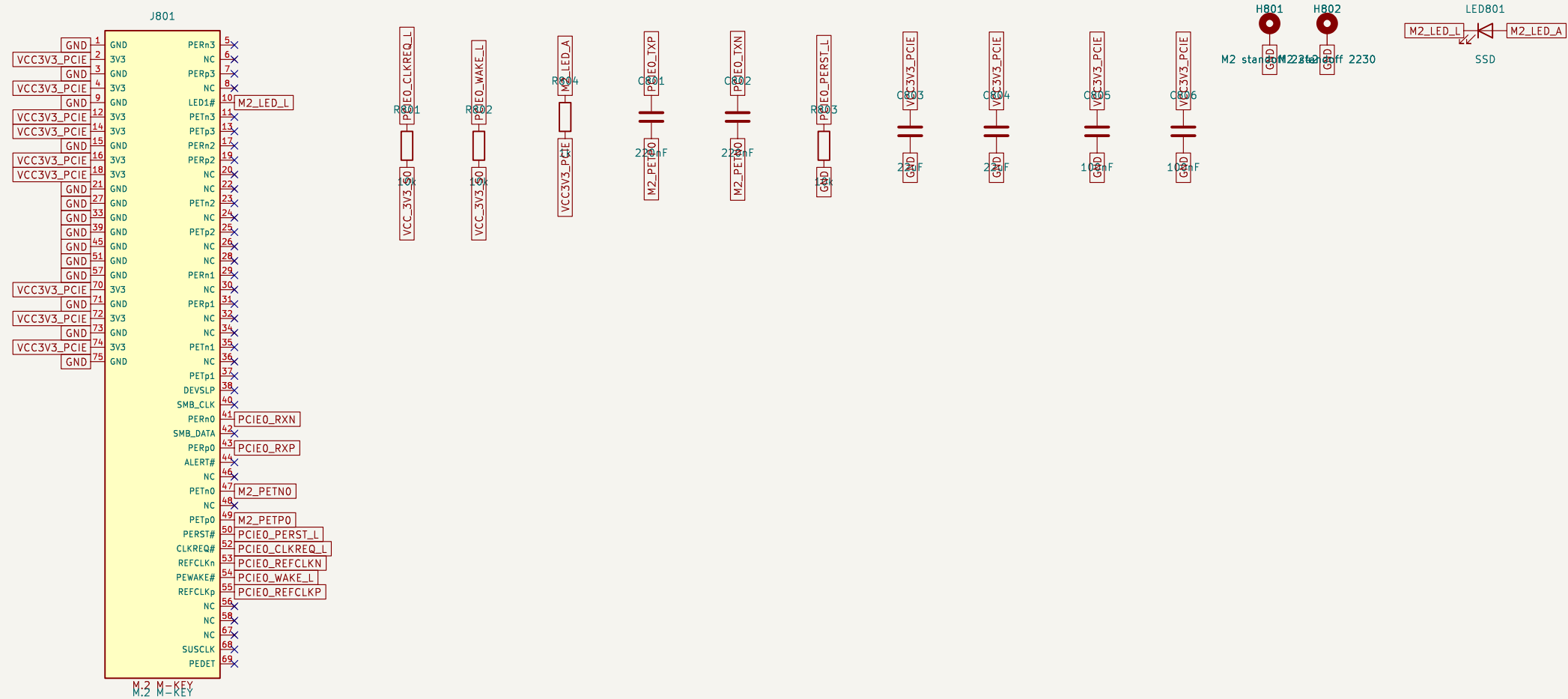
Generated by tools/gen_schematic.py from tools/design.py – edit the source, not this file
RK3576 SBC (LPDDR5 / eMMC / NVMe / HDMI 4K)
 Sheet: /05_soc_system/
 File: 05_soc_system.kicad_sch
Title: Clock, reset, boot, debug UART, keys, LEDs
 Size: A3 Date: 2026-10-04 **Rev: A0—draft**
 KiCad E.D.A. 9.0.9 Id: 6/13

[illegible]

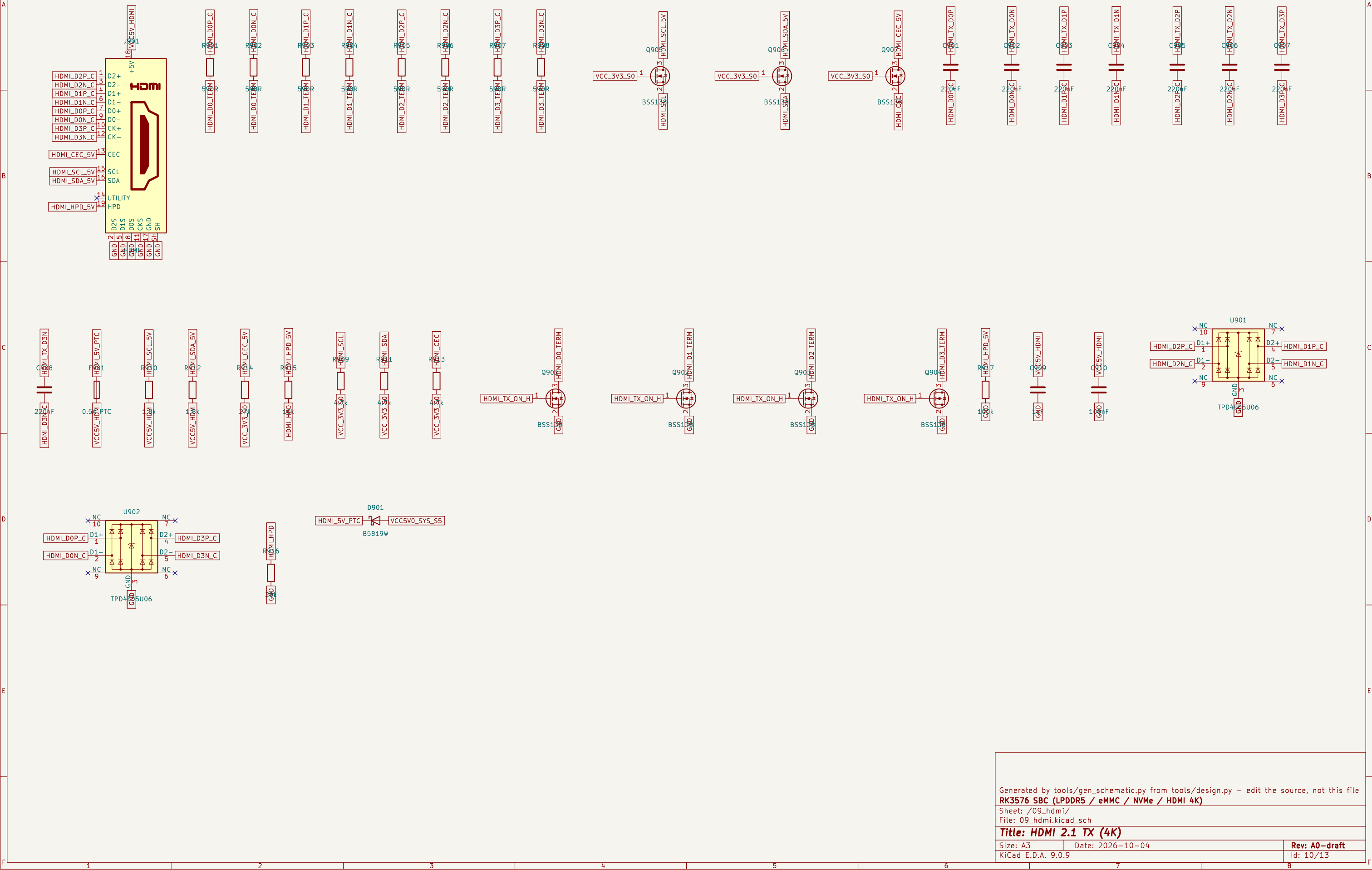
07_storage: eMMC 5.1 + microSD



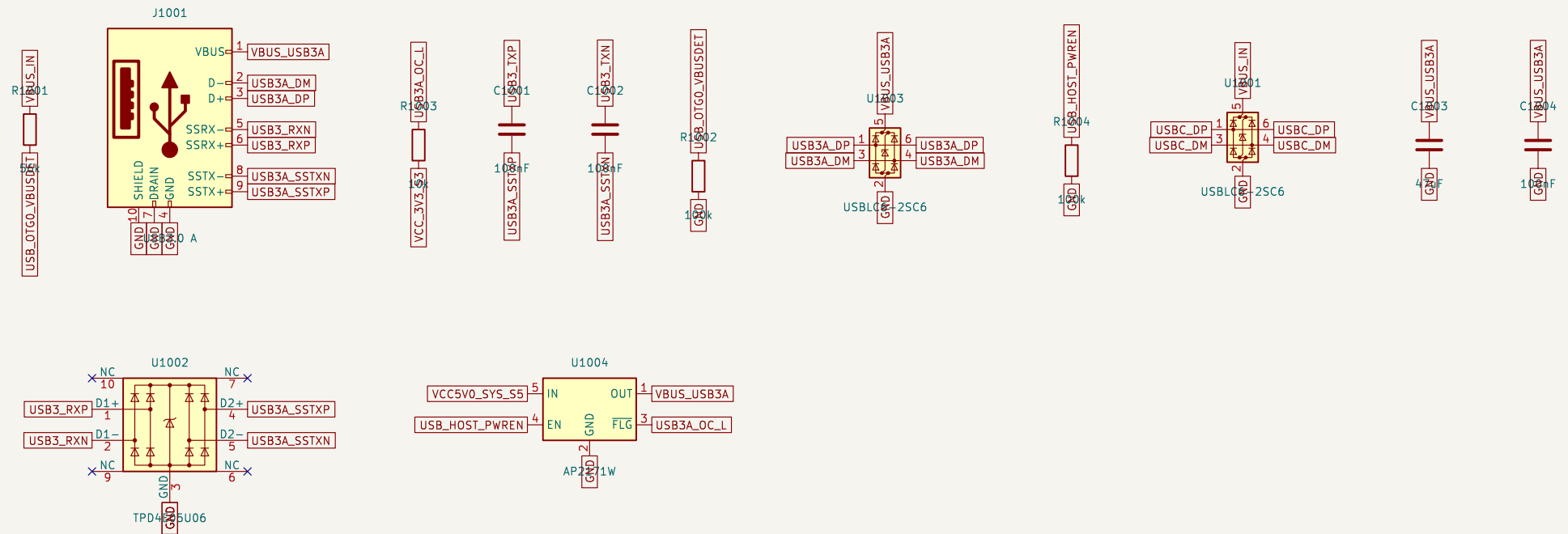
08_nvme: M.2 M-key NVMe (PCIe 2.1 x1)



09_hdmi: HDMI 2.1 TX (4K)



10_usb: USB-C OTG (maskrom) + USB 3.0 host



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RK3576 SBC (LPDDR5 / eMMC / NVMe / HDMI 4K)

Sheet: /10_usb/
File: 10_usb.kicad_sch

Title: USB-C OTG (maskrom) + USB 3.0 host

Size: A4	Date: 2026-10-04
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Rev: A0-draft

Size: A4	
KiCad E.D.A. 9.0.9	

Id: 11/13

11_soc_gpio: RK3576 GPIO banks

U401E

SD_D0	B21	GPIO2_A0/SDMMC0_D0	GPIO2_C0/VL_CIF_D5	A17
SD_D1	B25	GPIO2_A1/SDMMC0_D1	GPIO2_C1/VL_CIF_D4	A15
SD_D2	A25	GPIO2_A2/SDMMC0_D2	GPIO2_C2/VL_CIF_D3	D15
SD_D3	B24	GPIO2_A3/SDMMC0_D3	GPIO2_C3/VL_CIF_D2	A14
SD_CMD	A22	GPIO2_A4/SDMMC0_CMD	GPIO2_C4/VL_CIF_D1	A13
SD_CLK	B22	GPIO2_A5/SDMMC0_CLK	GPIO2_C5/VL_CIF_D0	C15
	B24	GPIO2_A6/VL_CIF_D15	GPIO2_C6/ETH1_TXD0_M0	A14
	B20	GPIO2_A7/VL_CIF_D14	GPIO2_C7/ETH1_TXD1_M0	B15
	B19	GPIO2_B0/VL_CIF_D13	GPIO2_D0/ETH1_TXCTL_M0	B19
	A18	GPIO2_B1/VL_CIF_D12	GPIO2_D1/ETH1_RXD0_M0	A16
	A11	GPIO2_B2/VL_CIF_D11	GPIO2_D2/CAM_CLK0_OUT_M1	B17
	B19	GPIO2_B3/VL_CIF_D10	GPIO2_D3/ETH1_RXCTL_M0	B18
	A16	GPIO2_B4/VL_CIF_D9	GPIO2_D4/ISP_PRELIGHT_TRIG_M0	B13
	C19	GPIO2_B5/VL_CIF_D8	GPIO2_D5/ISP_FLASH_TRIGOUT_M0	B15
	A21	GPIO2_B6/VL_CIF_D7	GPIO2_D6/CAM_CLK1_OUT_M1	D18
	B21	GPIO2_B7/VL_CIF_D6	GPIO2_D7/CAM_CLK2_OUT_M1	E15

RK3576 GPIO2
RK3576

U401F

GPIO3_A0/VL_CIF_HREF	GPIO3_C0/V0_LCDC_D11	E9
GPIO3_A1/VL_CIF_VSYNC	GPIO3_C1/V0_LCDC_D10	B10
GPIO3_A2/VL_CIF_CLK0	GPIO3_C2/V0_LCDC_D9	B9
GPIO3_A3/VL_CIF_CLK1	GPIO3_C3/V0_LCDC_D8	A8
GPIO3_A4/V0_LCDC_D23	GPIO3_C4/V0_LCDC_D7	D9
GPIO3_A5/V0_LCDC_D22	GPIO3_C5/V0_LCDC_D6	B8
GPIO3_A6/V0_LCDC_D21	GPIO3_C6/V0_LCDC_D5	D7
GPIO3_A7/V0_LCDC_D20	GPIO3_C7/V0_LCDC_D4	C7
GPIO3_B0/V0_LCDC_D19	GPIO3_D0/V0_LCDC_D3	C6
GPIO3_B1/V0_LCDC_D18	GPIO3_D1/V0_LCDC_D2	A1
GPIO3_B2/V0_LCDC_D17	GPIO3_D2/V0_LCDC_D1	A10
GPIO3_B3/V0_LCDC_D16	GPIO3_D3/V0_LCDC_D0	B12
GPIO3_B4/V0_LCDC_D15	GPIO3_D4/V0_LCDC_DEN	E12
GPIO3_B5/V0_LCDC_D14	GPIO3_D5/V0_LCDC_HSYNC	D10
GPIO3_B6/V0_LCDC_D13	GPIO3_D6/V0_LCDC_VSYNC	C10
GPIO3_B7/V0_LCDC_D12	GPIO3_D7/V0_LCDC_CLK	E7

RK3576 GPIO3
RK3576

U401C

GPIO0_A0/REF_CLK0_OUT	GPIO0_B7/I2C2_SCL_M0	W24
GPIO0_A1/TSADC_CTRL_M0	GPIO0_C0/I2C2_SDA_M0	B29
GPIO0_A2/CLK_32K_IN	GPIO0_C1/I2C0_SCL_M1	A28
GPIO0_A3/PWR_CTRL1	GPIO0_C2/I2C0_SDA_M1	V26
GPIO0_A4/PWR_CTRL2	GPIO0_C3/PDM0_CLK1_M0	W27
GPIO0_A5/PWR_CTRL3	GPIO0_C4/SAIO_MCLK_M1	A27
GPIO0_A6/PMIC_INT	GPIO0_C5/SAIO_SD00_M1	A21
GPIO0_A7/SDMMC0_DET_N	GPIO0_C6/SAIO_SCLK_M1	V21
GPIO0_B0/AUPLL_CLK_IN_M1	GPIO0_C7/SAIO_LRCK_M1	V23
GPIO0_B1/SPI2_MISO_M0	GPIO0_D0/SAIO_SDIO_M1	W23
GPIO0_B2/SPI2_CLK_M0	GPIO0_D1/SAIO_SDIO_M1	A28
GPIO0_B3/SPI2_MOSI_M0	GPIO0_D2/SAIO_SDIO_M1	V22
GPIO0_B4/REF_CLK1_OUT	GPIO0_D3/SAIO_SDIO_M1	A23
GPIO0_B5/REF_CLK2_OUT	GPIO0_D4/UART0_TX_M0	U21
GPIO0_B6/SDMMC0_PWREN	GPIO0_D5/UART0_RX_M0	A22

RK3576 GPIO0
RK3576

U401D

EMMC_DAT0	E26	GPIO1_A0/EMMC_D0	GPIO1_B7/ETH1_TXD2_M1	A27
EMMC_DAT1	E29	GPIO1_A1/EMMC_D1	GPIO1_C0/ETH1_TXD3_M1	B26
EMMC_DAT2	C2	GPIO1_A2/EMMC_D2	GPIO1_C1/ETH1_TXCLK_M1	B22
EMMC_DAT3	E26	GPIO1_A3/EMMC_D3	GPIO1_C2/ETH1_PPSCCLK_M1	B29
EMMC_DAT4	D2	GPIO1_A4/EMMC_D4	GPIO1_C3/ETH1_PPSTRIG_M1	C23
EMMC_DAT5	E26	GPIO1_A5/EMMC_D5	GPIO1_C4/ETH1_TXD0_M1	B23
EMMC_DAT6	D26	GPIO1_A6/EMMC_D6	GPIO1_C5/ETH1_TXD1_M1	B28
EMMC_DAT7	B2	GPIO1_A7/EMMC_D7	GPIO1_C6/ETH1_TXCTL_M1	B26
EMMC_CMD	E22	GPIO1_B0/EMMC_CMD	GPIO1_C7/ETH1_RXD0_M1	C22
EMMC_CLK	G29	GPIO1_B1/EMMC_CLK	GPIO1_D0/ETH1_RXD1_M1	C29
EMMC_DS	E29	GPIO1_B2/EMMC_STRB	GPIO1_D1/ETH1_RXCTL_M1	D22
EMMC_RSTN	D29	GPIO1_B3/EMMC_RSTN	GPIO1_D2/ETH1_MDC_M1	A24
	A26	GPIO1_B4/ETH1_RXD2_M1	GPIO1_D3/ETH1_MDIO_M1	C28
	B2	GPIO1_B5/ETH1_RXD3_M1	GPIO1_D4/ETH1_MCLK_M1	E21
	A22	GPIO1_B6/ETH1_RXCLK0_M1	GPIO1_D5/ETH_CLK1_25M_OUT_M1	E22

RK3576 GPIO1
RK3576

U401G

GPIO4_A0/SPDIF_RX0_M1	GPIO4_B4/SPDIF_RX0_M0	B8
GPIO4_A1/V0_POST_EMPTY	GPIO4_B5/SPDIF_TX0_M0	A8
GPIO4_A2/SAI1_MCLK_M0	GPIO4_C0/DSM_AUD_LP_M1	AK3
GPIO4_A3/SAI1_SCLK_M0	GPIO4_C1/DSM_AUD_LN_M1	A2
GPIO4_A4/SAI4_SCLK_M0	GPIO4_C2/DSM_AUD_RP_M1	A2
GPIO4_A5/SAI1_LRCK_M0	GPIO4_C3/DSM_AUD_RN_M1	A2
GPIO4_A6/SAI4_LRCK_M0	GPIO4_C4/ISP_PRELIGHT_TRIG_M1	A13
GPIO4_A7/SAI1_SD00_M0	GPIO4_C5/ISP_FLASH_TRIGOUT_M1	AK3
GPIO4_B0/SAI1_SD01_M0	GPIO4_C6/SAI4_SDL_M2	A2
GPIO4_B1/SAI1_SD02_M0	GPIO4_C7/SAI4_SCLK_M2	AJ1
GPIO4_B2/SAI1_SD03_M0	GPIO4_D0/UFS_RSTN	A02
GPIO4_B3/SAI1_SDIO_M0	GPIO4_D1/UFS_REFCLK	AK4

RK3576 GPIO4
RK3576

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RK3576 SBC (LPDDR5 / eMMC / NVMe / HDMI 4K)
Sheet: /11_soc_gpio/
File: 11_soc_gpio.kicad_sch
Title: RK3576 GPIO banks
Size: A4 Date: 2026-10-04 Rev: A0-draft
KiCad E.D.A. 9.0.9 Id: 12/13

Sheet: /11_soc_gpio/
File: 11_soc_gpio.kicad_sch

Size: A4	Date: 2026-10-04	Rev: A0-draft
KiCad E.D.A. 9.0.9		Id: 12/13

12_mechanical: Mounting holes, fiducials



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RK3576 SBC (LPDDR5 / eMMC / NVMe / HDMI 4K)

Sheet: /12_mechanical/
File: 12_mechanical.kicad_sch

Title: Mounting holes, fiducials

Size: A4	Date: 2026-10-04	Rev: A0-draft
KiCad E.D.A. 9.0.9		Id: 13/13